ANDRÉ ROGÉRIO DA SILVA FILHO

Software Development Consultant | Full-Stack • Autonomous AI Agents • Cloud Architecture

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SUMMARY

Software Development Consultant specializing in full-stack architecture, autonomous AI agents, and cloud infrastructure, with a strong engineering foundation established in corporate IT operations and systems. Proven track record designing and delivering mission-critical production platforms using TypeScript, Python, React, Next.js, Node.js, and PostgreSQL. At Grupo CBI, progressed from Full-Stack Developer to Software Development Consultant, leading enterprise software engineering, autonomous AI agent architecture (Google Gemini, MCP servers, advanced RAG), cloud infrastructure on GCP (Cloud Run, Cloud SQL, Secret Manager), and high-availability relational databases safeguarding sensitive citizen data across metropolitan concessions under strict LGPD compliance. In technological research (UniFacens/MCTI/Softex), ranked 8th nationwide, engineering industrial digital twins, real-time WebSocket telemetry, and cognitive memory architectures with 3D spatial interfaces.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, React 19/18, Next.js 15, Express, Fastify, PyTest, Playwright, Vitest, Three.js, React Three Fiber, WebGL/GLSL Shaders

Applied AI & Agents: Autonomous AI Agents (LangGraph, MCP Servers), LLMs (Google Gemini API), Vision-Language Models (Qwen2-VL via vLLM on AMD ROCm), PyTorch, Advanced RAG (Hybrid Search & Reranking), NetworkX Relational Graphs, Structured Outputs (JSON Schema), Zod

Cloud, DevOps & Data: Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager), Docker, CI/CD Pipelines, PostgreSQL (Row-Level Security), Redis, Supabase, Prisma ORM, Git

Enterprise, Security & Compliance: LGPD/GDPR Compliance, AES-256-GCM Encryption, IDOR Prevention, RFC 6962 Merkle Trees, Microsoft 365, Entra ID, Microsoft Graph API, Microsoft Intune, Certified AppSec Practitioner (CAP)

PROJECTS

CBI Operations Center | Enterprise ITSM & Fleet Telemetry Platform

Grupo CBI

TypeScript, React 19, Node.js, Express, PostgreSQL, Google Cloud Run, Gemini API, Microsoft Graph API, Intune

- Architected and built from scratch an enterprise ITSM platform managing assets, tickets, and operational workflows across infrastructure concessions, cutting projected TCO by 80% to 92%+ versus market solutions.
- Engineered a confined AI deflection engine using Google Gemini and contextual RAG retrieval, projecting operational savings of R$ 32K to R$ 114K/year with automated triage and 15 to 38 minutes saved per incident.
- Integrated real-time fleet compliance telemetry and identity lifecycle governance via Microsoft Graph API, Entra ID, and Intune, containerized on Google Cloud Run.

Cribum (AuditRx) | Multimodal Medical Audit & Fraud Graph

GitHub

AMD Developer Hackathon & Campus Mobile • Python, FastAPI, React 19, PyTorch, AMD ROCm, vLLM, Qwen2-VL, NetworkX, TypeScript

- Architected a local, privacy-first multi-agent system for prescription audit, LASA drug safety (ISMP/ANVISA), and healthcare fraud detection submitted to AMD Developer Hackathon and Campus Mobile.
- Integrated local Vision-Language Models (Qwen2-VL on vLLM) with uncertainty-aware multi-hypothesis extraction, clinical risk scoring (CCRI), and an RFC 6962 Merkle Tree custody chain for LGPD compliance.
- Engineered a relational multigraph anomaly detection engine with NetworkX and a comprehensive 36-test automated suite (22 Pytest unit/integration tests and 14 Playwright E2E browser scenarios).

Memory Lens | Cognitive Computing Platform & 3D Neural Interface

GitHub

TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Gemini API, Vitest

- Designed a cognitive memory platform modeling human retention, associative recall, and structured forgetting (Ebbinghaus curve), reducing token storage overhead by up to 80%.
- Engineered a 3D bioluminescent neural organism in React Three Fiber with custom GLSL shaders synchronized to system cognitive state via an event bus.
- Conducted formal academic research on semantic document organization under Prof. Dr. Fabrício Londero (UniFacens), targeting submission to the ACDSA international conference.

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura) 07/2026 – Present

Software Development Consultant | São Paulo, Brazil (Remote) (10/2026 – Present)

- Promoted to Software Development Consultant leading enterprise software engineering, internal platform architecture, and autonomous AI systems for mission-critical operations.
- Architect and manage cloud infrastructure on Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager) and relational databases, directly responsible for safeguarding sensitive citizen data across metropolitan public concessions in Brazil under strict LGPD compliance.
- Direct technical engagements and architecture alignment with enterprise platform vendors (including Google Cloud and Microsoft), ensuring rigorous security, zero-trust access, and infrastructure resilience.
- Conceived, architected, and deployed "Isa", an enterprise AI agent utilizing Google Gemini, autonomous tool execution via MCP Servers, contextual RAG retrieval, and deterministic JSON Schema validation with Zod.
- Conduct corporate technical enablement workshops and establish AI governance frameworks, training engineering and operational teams on secure LLM workflows.

Full-Stack Developer | São Paulo, Brazil (Remote) (07/2026 – 10/2026)

- Engineered full-stack web applications and internal tools using TypeScript, React 19, Next.js, Node.js, and PostgreSQL.
- Implemented secure API integrations, webhooks, and automated security monitoring pipelines connecting enterprise platforms with Microsoft Graph API and Intune telemetry.
- Developed custom automation scripts in Python and TypeScript to optimize cross-departmental operations and incident deflection, resulting in promotion to Software Development Consultant.

Farmamed Drogarias 10/2025 – 03/2026

IT Infrastructure Analyst | Sorocaba, Brazil

- Managed corporate IT infrastructure, identity governance, access provisioning, and endpoint compliance across Microsoft Entra ID, Microsoft Intune, and Active Directory across multi-site retail branches.
- Developed Python automation scripts and data consolidation pipelines for log analytics, network diagnostics, and operational synchronization.
- Monitored mission-critical services and resolved high-priority technical incidents under strict operational SLAs.

Wobben Windpower 11/2023 – 04/2025

Software & IT Systems Intern | Sorocaba, Brazil

- Selected for the global Microsoft Copilot Flight Crew program, evaluating generative AI enterprise features and real-world adoption workflows.
- Developed internal automation scripts and integration routines for SAP ERP, Active Directory, and Microsoft 365 in a heavy industrial manufacturing environment.
- Provided tier-2 technical resolution for corporate service desk incidents via ServiceNow under strict SLA benchmarks.

EDUCATION

Higher Technology Degree in Systems Analysis and Development (TADS) 2026

Centro Universitário SENAC, Brazil | Official Degree Average: 8.20 / 10.0 (US GPA: 3.38 / 4.0)

TECHNOLOGY RESIDENCY & APPLIED RESEARCH

ICT Resident Researcher (Digital Twins, XR & 5G) 10/2026 – 04/2027

UniFacens | Softex | Ministry of Science, Technology and Innovation (MCTI) | ICT Residency 51 (PPI)

- Ranked **8th nationwide out of 300+ candidates** in Phase 1: an intensive 360-hour specialization across AI, IoT, Extended Reality (XR), and 5G with a **4.00 / 4.0 GPA** (9.91 / 10.0 average, 10/10 in 10 modules), securing one of only 20 federal research fellowships awarded.
- Selected for Phase 2: a 6-month Technological Residency conducting applied research and enterprise PoCs across **Digital Shadow, Computer Vision, and Systems Integration**.
- Served as team leader across all 4 collaborative capstone projects and architected FundiVR, an industrial digital twin simulator for 700°C+ metallurgy operations with real-time WebSocket telemetry, defended and approved with 10/10.

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader | Azure Well-Architected Framework & Cloud Architecture (Fast Lane / Microsoft) | Certified AppSec Practitioner (CAP) | Cisco CCNA 1 & Network Security

Languages: English (C1+ / Fluent, 2-Year Full Immersion at Happy & Co., Speaking: 4.5/5, Listening: 4.4/5) | Spanish (Intermediate / Technical) | Portuguese (Native)

Teaching & Mentoring: Former English Teacher & Mentor at Clube Poliglota Sorocaba, conducting cross-cultural speaking sessions for local and international attendees (including Latin America and Southeast Asia).